

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 15 – Review Version 1

Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury. They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.

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Topic 16 – Review Version 1

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.

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Topic 17 – Review Version 1

High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects. Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent.

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Topic 18 – Review Version 1

It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy. Diabetic retinopathy is the leading cause of new blindness in working-age adults.

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Topic 19 – Review Version 1

Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support. It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients

Topic 10 – Review Version 1

Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.. Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia.

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Topic 11 – Review Version 1

COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure. Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.

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Topic 12 – Review Version 1

WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.. Physical inactivity contributes to 3.2 million deaths annually globally.

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Topic 13 – Review Version 1

Medication reconciliation is the process of creating the most accurate possible list of all medications a patient is taking, including dose, frequency, and route, and comparing it against the physician's admission, transfer, or discharge orders to identify and resolve discrepancies. It reduces medication errors at care transitions, which are responsible for approximately 60 percent of serious medication errors in hospital settings. Discrepancies found during reconciliation include omitted medications, wrong doses, duplicated medications, and medications continued that should have been stopped. All newly hospitalised patients should have medication reconciliation performed by a pharmacist within 24 hours of admission.

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Topic 14 – Review Version 1

Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.. Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor.

recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.

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Topic 6 – Review Version 1

Oedema represents excess fluid accumulation in the interstitial compartment and is classified by distribution and cause. Bilateral leg oedema suggests systemic causes including heart failure (elevated venous pressure with sodium retention), nephrotic syndrome (protein loss reducing oncotic pressure), liver cirrhosis (reduced albumin synthesis and portal hypertension), or hypothyroidism causing myxoedema. Unilateral leg oedema suggests local causes including deep vein thrombosis, venous insufficiency, lymphoedema, or infection. Investigation follows the distribution: bilateral oedema warrants cardiac, renal, liver, and thyroid assessment; unilateral oedema warrants duplex ultrasound to exclude DVT.

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Topic 7 – Review Version 1

The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials. It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.

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Topic 8 – Review Version 1

Urinary tract infections are classified as uncomplicated (affecting otherwise healthy, non-pregnant women with normal urinary tracts) or complicated (occurring in men, pregnant women, those with structural abnormalities, immunosuppression, or catheter-related). *Escherichia coli* causes approximately 80 percent of community-acquired UTIs. Uncomplicated lower UTI in women is treated with a 3-day course of trimethoprim or nitrofurantoin based on local resistance patterns. Recurrent UTIs (three or more per year) warrant investigation for predisposing factors and may be managed with low-dose prophylactic antibiotics, post-coital prophylaxis, or self-start therapy.

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Topic 9 – Review Version 1

These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients. GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control.

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Expanded Medical Reference

Topic 1 – Review Version 1

Patient-centred care is an approach that respects and responds to individual patient preferences, needs, and values, ensuring that patient values guide all clinical decisions. Its core dimensions include respect for patient values and expressed needs, emotional support and relief of fear and anxiety, information and communication in accessible language, coordination and integration of care across settings, physical comfort and pain management, involvement of family and friends as appropriate, and access to care and continuity. Evidence consistently shows that patient-centred care improves patient satisfaction, treatment adherence, health outcomes, and reduces unnecessary investigations and readmissions.

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Topic 2 – Review Version 1

Prediabetes is characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent. Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.

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Topic 3 – Review Version 1

Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure. Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.

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Topic 4 – Review Version 1

Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis. They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.

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Topic 5 – Review Version 1

SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria. Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now